

Problem Set 77

Collection of Problems

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Problem 3

Let $L = \lim_{x \rightarrow 0} \frac{a - \sqrt{a - x^2} - \frac{x^2}{4}}{x^4}$ be a finite limit. Find the values of $a(a > 0)$ and L .

$$\begin{aligned}
 L &= \lim_{x \rightarrow 0} \frac{a - \sqrt{a - x^2} - \frac{x^2}{4}}{x^4} \quad \left[\begin{array}{c} 0 \\ 0 \end{array} \right] \\
 &\begin{cases} \lim_{x \rightarrow 0} a - \sqrt{a - x^2} - \frac{x^2}{4} = 0 \\ \lim_{x \rightarrow 0} x^4 = 0 \end{cases} \\
 L &\stackrel{\left[\begin{array}{c} 0 \\ 0 \end{array} \right]}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0} \frac{\frac{x}{\sqrt{a - x^2}} - \frac{x}{2}}{4x^3} = \lim_{x \rightarrow 0} \frac{\frac{1}{\sqrt{a - x^2}} - \frac{1}{2}}{4x^2} \\
 &\begin{cases} \lim_{x \rightarrow 0} 4x^2 = 0 \\ \lim_{x \rightarrow 0} \frac{1}{\sqrt{a - x^2}} - \frac{1}{2} = 0 \implies a = 2 \end{cases} \\
 L &\stackrel{\left[\begin{array}{c} 0 \\ 0 \end{array} \right]}{\underset{\text{L'H}}{=}} \lim_{x \rightarrow 0} \frac{-\frac{1}{2} \cdot \frac{-2x}{(2 - x^2)^{3/2}}}{8x} = \lim_{x \rightarrow 0} \frac{1}{8 \cdot 2^{3/2}} = \frac{\sqrt{2}}{32} \\
 &\boxed{a = 2, L = \frac{\sqrt{2}}{32}}
 \end{aligned}$$

Problem 4

Let $f(x) = (1 - x)^2(\sin x)^2 + x^2$ for all $x \in \mathbb{R}$. Consider the statements:

P: There exists some $x \in \mathbb{R}$ such that

$$f(x) + 2x = 2(1 + x^2)$$

Q: There exists some $x \in \mathbb{R}$ such that

$$2f(x) + 1 = 2x(1 + x)$$

Then

- (A) both P and Q are true (B) P is true and Q is false
 (C) P is false and Q is true (D) both P and Q are false

Justify your answer.

$$P: f(x) = 2x^2 - 2x + 2$$

$$Q: f(x) = x^2 + x - \frac{1}{2}$$

For P:

$$(1-x)^2(\sin x)^2 + x^2 = 2x^2 - 2x + 2$$

$$(1-x)^2 \sin^2 x = x^2 - 2x + 2$$

$$(1-x)^2 \sin^2 x = (1-x)^2 + 1$$

$$(1-x)^2 \cos^2 x = -1$$

As no real x satisfies the above equality, P is false. For Q:

$$h(x) = 2f(x) + 1 - 2x(1+x)$$

$$h(0) = 2f(0) + 1 - 0 = 1$$

$$h(1) = 2f(1) + 1 - 2 \cdot 2 = -2$$

By the Intermediate Value Theorem, $h(x)$ holds a solution on $(0, 1)$. Hence, Q is true.

$(C) \text{ P is false and Q is true}$

Problem 5

Discuss continuity and differentiability

$$f(x) = \begin{cases} -x - \frac{\pi}{2} & \text{if } x \leq -\frac{\pi}{2} \\ -\cos x & \text{if } -\frac{\pi}{2} < x \leq 0 \\ x - 1 & \text{if } 0 < x \leq 1 \\ \ln x & \text{if } x > 1 \end{cases}$$

$$f'(x) = \begin{cases} -1 & \text{if } x \leq -\frac{\pi}{2} \\ \sin x & \text{if } -\frac{\pi}{2} < x \leq 0 \\ 1 & \text{if } 0 < x \leq 1 \\ \frac{1}{x} & \text{if } x > 1 \end{cases}$$

At $x = -\frac{\pi}{2}$:

$$f(-\frac{\pi}{2}^-) = 0, f(-\frac{\pi}{2}^+) = 0, f(-\frac{\pi}{2}) = 0$$

$$f(-\frac{\pi}{2}^-) = f(-\frac{\pi}{2}^+) = f(-\frac{\pi}{2})$$

$$f'(-\frac{\pi}{2}^-) = -1, f'(-\frac{\pi}{2}^+) = -1$$

$$f'(-\frac{\pi^-}{2}) = f'(-\frac{\pi^+}{2}) \implies f'(-\frac{\pi}{2}) = -1$$

$f(x)$ is continuous and differentiable at $x = -\frac{\pi}{2}$. At $x = 0$:

$$f(0^-) = -1, f(0^+) = -1, f(0) = -1$$

$$f(0^-) = f(0^+) = f(0)$$

$$f'(0^-) = 0, f'(0^+) = 1$$

$$f'(0^-) \neq f'(0^+) \implies f'(0) \text{ DNE}$$

$f(x)$ is continuous but not differentiable at $x = 0$. At $x = 1$:

$$f(1^-) = 0, f(1^+) = 0, f(1) = 0$$

$$f(1^-) = f(1^+) = f(1)$$

$$f'(1^-) = 1, f'(1^+) = 1$$

$$f'(1^-) = f'(1^+) \implies f'(1) = 1$$

$f(x)$ is continuous and differentiable at $x = 1$.

Problem 6

Find the domain of the derivative function

$$f(x) = \begin{cases} \arctan x & \text{if } |x| \leq 1 \\ \frac{1}{2}(|x| - 1) & \text{if } |x| > 1 \end{cases}$$

$$f'(x) = \begin{cases} \frac{1}{1+x^2} & \text{if } |x| \leq 1 \\ \frac{1}{2} \operatorname{sgn}(x) & \text{if } |x| > 1 \end{cases}$$

At $x = -1$:

$$f'(-1^-) = -\frac{1}{2}, f'(-1^+) = \frac{1}{2}$$

$$f'(-1^-) \neq f'(-1^+) \implies f'(-1) \text{ DNE}$$

$f(x)$ is not differentiable at $x = -1$. At $x = 1$:

$$f'(1^-) = \frac{1}{2}, f'(1^+) = \frac{1}{2}$$

$$f'(1^-) = f'(1^+) \implies f'(1) = \frac{1}{2}$$

$f(x)$ is differentiable at $x = 1$.

$$\boxed{D_{f'} = (-\infty, -1) \cup (-1, \infty)}$$

Problem 7

Let $g(x) = \ln f(x)$ where $f(x)$ is twice differentiable positive function on $(0, \infty)$ such that $f(x+1) = xf(x)$. Find

$$g''\left(x + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right)$$

(in terms of N), where $x = 1, 2, 3, \dots, N$.

$$\ln f(x+1) = \ln x + \ln f(x)$$

$$g(x+1) - g(x) = \ln x$$

$$g'(x+1) - g'(x) = \frac{1}{x}$$

$$g''(x+1) - g''(x) = -\frac{1}{x^2}$$

$$x = \frac{1}{2} \implies g''\left(\frac{3}{2}\right) - g''\left(\frac{1}{2}\right) = -4$$

$$x = \frac{3}{2} \implies g''\left(\frac{5}{2}\right) - g''\left(\frac{3}{2}\right) = -\frac{4}{9}$$

$$x = \frac{5}{2} \implies g''\left(\frac{7}{2}\right) - g''\left(\frac{5}{2}\right) = -\frac{4}{25}$$

$$g''\left(N + \frac{1}{2}\right) - g''\left(\frac{1}{2}\right) = -4 \sum_{k=1}^N \frac{1}{(2k-1)^2}$$

Problem 8

Let $f(\theta) = \sin\left(\arctan\left(\frac{\sin \theta}{\sqrt{\cos 3\theta}}\right)\right)$, where $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$. Find $\frac{d}{d(\tan \theta)}(f(\theta))$.

$$\begin{aligned} & \frac{d}{d(\tan \theta)}(f(\theta)) \\ &= \frac{\frac{df}{d\theta}}{\frac{d(\tan \theta)}{d\theta}} \\ &= \frac{\cos\left(\arctan\left(\frac{\sin \theta}{\sqrt{\cos 3\theta}}\right)\right) \left(\frac{1}{1 + \frac{\sin^2 \theta}{\cos 3\theta}}\right) \left(\frac{\sin \theta}{\sqrt{\cos 3\theta}}\right)'}{\sec^2 \theta} \\ &= \cos\left(\arctan\left(\frac{\sin \theta}{\sqrt{\cos 3\theta}}\right)\right) \left(\frac{1}{\sec^2 \theta (1 + \frac{\sin^2 \theta}{\cos 3\theta})}\right) \frac{\cos \theta \sqrt{\cos 3\theta} + \sin \theta \frac{3 \sin 3\theta}{2\sqrt{\cos 3\theta}}}{\cos 3\theta} \\ &= \frac{\sqrt{\cos 3\theta}}{\sqrt{\cos 3\theta + \sin^2 \theta}} \left(\frac{\cos^2 \theta \cos 3\theta}{\cos 3\theta + \sin^2 \theta}\right) \frac{\cos \theta \sqrt{\cos 3\theta} + \sin \theta \frac{3 \sin 3\theta}{2\sqrt{\cos 3\theta}}}{\cos 3\theta} \\ &= \boxed{\frac{\cos^2 \theta (2 \cos \theta \cos 3\theta + 3 \sin \theta \sin 3\theta)}{2(\cos 3\theta + \sin^2 \theta)^{\frac{3}{2}}}} \end{aligned}$$